

Why BirdDog Play (not a laptop + VLC) for theatre playback

Each theatre's screen is driven by a **BirdDog Play** — a dedicated, purpose-built NDI/SRT decoder — rather than a general-purpose laptop running a media player. This documents the reasoning, alongside the closest realistic alternative.

Zero-config in the theatre

BirdDog Play needs no on-site setup beyond physical connection: plug it into the theatre screen and the network, point it at a source, done. No OS to patch, no application to configure per-session, nothing for on-site staff to babysit beyond power and a cable. That matters at this scale — 12 theatres, unattended for most of a live event.

Protocol support: SRT primary, NDI backup, both natively

- **SRT** — the primary path in this design (see `docs/streaming-flow.md`) is bandwidth-efficient by design (long-GOP H.264, see `docs/bandwidth-analysis.md` for real figures). In this design it originates from VMix via the mothership's Restreamer, but the protocol keeps options open: the ATEM Mini Extreme ISO also has a **built-in hardware streaming engine** that can stream SRT directly over Ethernet with no computer involved (Blackmagic tech specs), unused here but available as a fallback origin. BirdDog Play receives either without caring which one is sending.
- **NDI** — the backup path (BirdDog Central → BirdDog Play, via the NDI Discovery Server — see `docs/streaming-flow.md`) is handled natively too, with no separate plugin or configuration step.

Remote control from the mothership

None of these require physically visiting a theatre — and the division of labour is deliberate, not three redundant tools:

- **Flock** — **the fleet-management tool in this design**: LAN discovery, tag-based grouping, full BirdUI-parity settings per device, and batch edits across a group (Rust + Docker). Purpose-built for managing many BirdDog Play units as one fleet rather than 12 individual devices, which is precisely this deployment's shape.
- **BirdUI** — each unit's own web admin, reachable directly, as the per-unit fallback (already the mechanism used to configure the NDI Discovery Server setting — `docs/streaming-flow.md`).
- **BirdDog Central** — retained in the design as a mothership VM (`docs/topology.md`) **for the NDI backup path's sender side, not as a management layer** — Flock covers the management job on its own. Central's routing/control console exists as a bonus, not a dependency.

Comparison: BirdDog Play vs. laptop + VLC

The realistic alternative is a laptop in each theatre running VLC, pointed at the same SRT/ NDI sources.

	BirdDog Play	Laptop + VLC
Setup	Plug in, done	Install/patch OS, install VLC, configure stream URL per source
SRT	Native, purpose-built decode	Native from VLC 3+ (SRT CookBook) — the one protocol VLC handles cleanly
NDI	Native, purpose-built decode	Requires a separate third-party plugin (NDI for VLC); reliably <i>outputting</i> VLC's playback as NDI is well-documented, but consistently <i>receiving and playing</i> an NDI source as input is far less so
Remote management	Flock (fleet-wide) with per-unit BirdUI — purpose-built for this device	Nothing built in — would need RDP/VNC/TeamViewer bolted on, none of it purpose-built for stream monitoring/control
Unattended reliability	Broadcast-grade embedded device, designed to run 24/7 unattended	General-purpose OS: updates, driver issues, VLC hangs on a dropped stream sometimes need a manual restart
Recovery from a dropped stream	Automatic reconnect	Not guaranteed — depends on VLC settings, may need manual intervention
Footprint/power	Small, low power, unobtrusive	Full laptop: bigger, hotter, more power, another thing that can be bumped/closed/updated mid-show
Attack surface	Purpose-built embedded device	Full general-purpose OS — much larger surface
Cost per unit	Lower, purpose-built	A laptop is not free even if "already owned" — it's now unavailable for anything else, and carries more ongoing maintenance

Recommendation

BirdDog Play, as already implemented throughout this design. The laptop + VLC path is real and technically works for SRT specifically, but loses on every axis that matters for 12 unattended theatres running for the duration of a live event: zero-config deployment, clean NDI support, purpose-built fleet management (Flock, with per-unit BirdUI), and broadcast-grade reliability without needing on-site intervention when something drops.